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Statistics Research Project Title

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The work contained in this thesis is my own work unless otherwise stated.

Signed: Kateryna KAPLUNOVA

Date: DATE

Acknowledgements

In the acknowledgments, you should account for anyone who supported you during your research. This could include family, your research peer group, other students, research group members, your supervisor, data providers, external partners, any computing service used, any financial support.

If your research peer group, other students, research group members, or your supervisors made substantive contributions towards your report – e.g. they identified suitable methods, provided code or contributed a sub-analysis – you must acknowledge their precise contribution here.

Abstract

Abstracts should be no longer than 300 words, and similar in style to a general science or statistics research paper.

General tips. Your research report will be judged on the quality of the work it describes according to the Marking Criteria shared with you. When writing your research report you may assume that the reader is familiar with the four core modules of the MSc in Statistics. Any additional other material should be explained to the reader, providing suitable references. See Section D of the supplementary material for more details. For example, you should avoid writing up in detail statistical methods discussed in one of the four core modules, but instead you should briefly present such methods and add appropriate references. More advanced methods or methods discussed in one of the elective modules should be described in more detail and referenced appropriately.

Your report must be presented within this template and be at most 35 pages in length, as counted by the Arabic page numerals (1,2,3...). This limit includes all figures and tables presented in the main text, excluding the Endmatter and References sections. This is approximately equivalent to a limit of 15,000 words. Shorter submissions may be graded highly, while excess length disproportionate to the content may be penalised.

Before submitting the research project, make sure you read the report in its entirety. There should be no half-finished sentences. All mathematical symbols should be defined. Use a spell-checker.

You will have to submit your report electronically in PDF format through the virtual learning environment (Blackboard). Please name this file in the following format: “CID-Surname-Firstname-Report.pdf”. Your report will be checked for plagiarism via online plagiarism detection services (e.g. Turnitin).

Note the report submission deadline is a very hard deadline since the assessment process has then to be completed on a very short timescale.

1. Introduction

Tips for the Introduction. This is where you describe the topic and the research objectives of your project.

You should attempt to set your work in the context of other work previously done in the field. Convey the background of your research referencing earlier work as appropriate. Define core terms. Include background and context, and focus as appropriate on the wider statistical context. Aim to demonstrate that you can confidently describe your project within a broader context of statistical research, as well as scientific research, that goes well beyond the more narrow topic of your project.

The introduction needs to demonstrate that you are aware of what you are doing, and how it relates to other work that should be properly referenced.

Clearly point out your main contributions at the end of the introduction, and provide an overview how the rest of your report is structured and links together.

Aim for approximately 1.5-3 pages, similar in style to a general science or statistics research paper.¹

2. Prior Work and Theoretical Foundations

2.1. Concepts and Importance of Anomaly Detection in Streaming Data

Anomaly detection refers to the task of finding patterns in data that do not conform to expected behaviour ([Chandola et al., 2009](#)). Such unusual patterns, or "anomalies", are crucial to detect in many domains since they can indicate significant, actionable information. For example, an anomalous network traffic pattern might signal a cyber-intrusion. Identifying anomalies helps uncover rare but important events that merit attention.

Anomaly detection in streaming data is present in many domains, ranging from health-care to environmental monitoring. Unlike static datasets, data streams carry unique challenges: (a) data arrive rapidly and (b) underlying patterns may change over time.

¹Tip: If you choose to use footnotes, do so sparingly.

The latter forms a key difficulty, as most traditional anomaly detection models are not designed to handle non-stationary data streams (Iglesias Vázquez et al., 2023).

2.2. Machine Learning Methods for Streaming Anomaly Detection

Modern anomaly detection heavily relies on machine learning (ML) techniques, especially unsupervised and semi-supervised learning, to flag anomalies without explicit labels. A wide range of ML-based methods have been applied, and many are widely used in industry due to their ability to learn complex patterns from data.

Classic approaches include distance- or density-based algorithms like the k-Nearest Neighbours outlier method and Local Outlier Factor (LOF). LOF, introduced by Breunig et al. (2000), computes an "outlier score" based on the local density of data points, where truly anomalous points have substantially lower density than their neighbours. As noted by Campos et al. (2016), LOF remains one of the most reliable options as a general-purpose outlier detection technique. While initially designed for static datasets, many such models have been adapted for streaming use through techniques like sliding windows or incremental updates.

Another major category is ensemble methods, which build multiple models and combine their results to improve robustness. A notable example is the Isolation Forest (Liu et al., 2008), which uses an ensemble of random binary trees to isolate observations - anomalies are typically isolated faster than normal points. In the streaming context, companies like Amazon have deployed a related tree-ensemble method called Random Cut Forest (RCF), initially introduced by Guha et al. (2016). RCF estimates anomaly scores in real time based on how new points alter the structure of a tree-based ensemble, making it both scalable and suitable for high-dimensional data.

In recent years, deep learning has also become prominent in anomaly detection. Neural network models can learn latent representations of "normal" behaviour and detect deviations; for readers interested in these, we refer to the works of Zhai et al. (2016) and Zong et al. (2018), which provide comprehensive treatments of unsupervised anomaly detection using neural methods.

2.3. Current Challenges in Streaming Anomaly Detection

Although machine learning models are central to most modern anomaly detection systems, they continue to face several persistent challenges. Most models typically operate under the assumption of a stable data distribution, making them ill-suited for non-stationary data streams, unless the models are frequently retrained or updated. Another key limitation is explainability: complex models such as ensembles and deep networks often act as black boxes, offering little to no insight into why an instance is flagged as anomalous (Li et al., 2023). Moreover, most ML detectors rely on heuristic definitions of anomalies without probabilistic guarantees or principled thresholding. Combined with the scarcity of real-time labels and the difficulty of evaluating rare events, these factors render streaming anomaly detection a complex and open research problem.

In contrast to these opaque models, Extreme Value Theory (EVT) can provide a mathematically grounded approach for modelling and detecting rare events in data streams, with explicit control over false alarm rates, statistically interpretable thresholds and minimal distributional assumptions.

2.4. Extreme Value Theory for Anomaly Detection

Extreme Value Theory (EVT) provides a statistical framework for modelling the behaviour of rare, extreme events in the tails of distributions, making it particularly suitable for anomaly detection. Unlike most machine learning methods, EVT offers a mathematically principled and distribution-agnostic approach for estimating the probability of extremes. Two foundational limit theorems form the basis of EVT-based anomaly detection presented below.

2.4.1. Fisher-Tippett-Gnedenko Theorem (GEV)

Theorem 2.1 (Fisher–Tippett–Gnedenko). Let $X_1, \dots, X_n \sim F$ i.i.d., and define $M_n = \max(X_1, \dots, X_n)$. If sequences $a_n > 0$, $b_n \in \mathbb{R}$ exist such that

$$\frac{M_n - b_n}{a_n} \xrightarrow{d} G_\gamma(x), \quad (1)$$

then the limiting distribution $G_\gamma(x)$ belongs to the **Generalised Extreme Value (GEV)** family:

$$G_\gamma(x) = \exp \left\{ - \left(1 + \gamma \frac{x - \mu}{\sigma} \right)^{-1/\gamma} \right\}, \quad 1 + \gamma \frac{x - \mu}{\sigma} > 0, \quad (2)$$

where $\mu \in \mathbb{R}$, $\sigma > 0$ and $\gamma \in \mathbb{R}$ are location, scale and shape parameters, respectively.

Theorem 2.1 demonstrates the universal limit behaviour of maxima, which is largely independent of the base distribution F , under mild regularity conditions (Gnedenko, 1943). This theoretical property underpins the distribution-agnostic nature of EVT-based anomaly detection.

2.4.2. Pickands–Balkema–de Haan Theorem (GPD)

Theorem 2.2 (Pickands–Balkema–de Haan). Given a high threshold u , the distribution of exceedances $Y = X - u \mid X > u$ converges to the **Generalised Pareto Distribution (GPD)**:

$$\mathbb{P}(X - u \leq y \mid X > u) \approx G_{\gamma, \sigma}(y), \quad (3)$$

with the GPD defined as

$$G_{\gamma, \sigma}(y) = \begin{cases} 1 - (1 + \gamma \frac{y}{\sigma})^{-1/\gamma}, & \gamma \neq 0, \\ 1 - \exp(-\frac{y}{\sigma}), & \gamma = 0, \end{cases} \quad y \geq 0, \quad 1 + \gamma \frac{y}{\sigma} > 0, \quad (4)$$

where $\sigma > 0$ and $\gamma \in \mathbb{R}$.

Theorem 2.2 forms the foundation of the **Peaks Over Threshold (POT)** method, which involves modelling the distribution of threshold exceedances using the GPD (Balkema and de Haan, 1974). This approach provides the theoretical basis for anomaly detection via principled estimation of extreme quantiles.

, pickands1975statistical, leadbetter1991extremes

2.4.3. Quantile Estimation under GPD

Let N_u be the number of exceedances over threshold u , out of n observations. Given an estimated GPD, a high quantile z_q satisfying $\mathbb{P}(X > z_q) = q$ can be estimated as:

$$z_q \approx u + \frac{\hat{\sigma}}{\hat{\gamma}} \left(\left(\frac{N_u}{nq} \right)^{\hat{\gamma}} - 1 \right), \quad \hat{\gamma} \neq 0.$$

In POT-based anomaly detection, setting a q allows for direct control of how often false alarms occur, while the corresponding threshold z_q serves as a principled, interpretable detection rule. Taken together, these features provide a significant advantage over the manually set thresholds typically used in ML-based detectors.

Despite its strong theoretical rigour, EVT relies on several assumptions for valid application. First, the data (especially exceedances) should be approximately i.i.d. or at least weakly dependent, a condition that underlies both the GEV and POT limit theorems. Second, the threshold u must be carefully calibrated: too high yields high variance through low sample size; too low introduces bias by risking including non-tail behaviour. Finally, EVT assumes tail stationarity.

These conditions are easily violated in most real-world streaming applications; the presence of autocorrelation, regime shifts, or evolving system dynamics undermines the reliability of POT-based methods. Addressing these limitations for non-stationary data is the central focus of this thesis.

2.5. POT for Streaming Anomaly Detection

Classical POT methods work well on static, stationary data, for which the discussed assumptions hold; however, real streaming data from open, dynamic environments often violates those. Consequently, researchers have developed streaming POT algorithms to adapt threshold models online. Notably, [Siffer et al. \(2017\)](#) introduced the SPOT and DSPOT algorithms that adjust anomaly thresholds on the fly, requiring only a user-specified risk level q (i.e. the desired false positive alarm rate) instead of a fixed threshold.

2.5.1. SPOT (Streaming Peaks-Over-Threshold)

SPOT sequentially updates an extreme-value threshold to flag anomalies in a univariate stream without assuming a fixed data distribution. First, an initial calibration batch of size n is used to fit a POT model and set two thresholds: an initial threshold t and an

“alarm” threshold z_q corresponding to the desired risk level q . Once calibrated, each new observation X_i is classified into one of three cases:

1. **Normal:** if $X_i < t$, the point is considered within the typical range.
2. **Peak (Extreme but not Anomaly):** if $t \leq X_i < z_q$, the point is an extreme event; it is used to update the GPD model, but is not flagged as an anomaly.
3. **Anomaly:** if $X_i \geq z_q$, the point is flagged as anomalous; it is *not* used to update the GPD.

Siffer et al. (2017) report that SPOT is robust to its hyperparameters. The initial batch size n mainly provides a starting estimate and, since we refit GPD as new peaks come in, does not significantly affect the long-term results. The authors also report that the risk parameter q can be chosen within a range of 10^{-5} to 10^{-3} without dramatic changes in detection performance. In practice, one would pick q based on an acceptable false-alarm rate, and SPOT will automatically adjust z_q to meet that target level on average. Regarding the initial threshold t , the authors fix its value at the 98th empirical percentile of the calibration data. While a theoretically justified threshold could be derived based on asymptotic EVT arguments (e.g., via the stability of exceedances), the fixed $t = 0.98$ was found to yield better empirical results.

2.5.2. DSPOT (Drift Streaming POT)

To release the stationarity assumption, DSPOT incorporates a drift correction through a preprocessing step that aims to transform the input series into approximately stationary residuals. Specifically, each new value X_i is transformed as follows:

$$X'_i = X_i - M_i$$

where M_i is a *local mean estimate* representing the baseline at time i . This estimate is computed via a *sliding window* of size d over the most recent non-anomalous observations. The aim is to centre the series around a local zero-mean, such that the residuals X'_i conform more closely to the assumptions required by EVT tail modelling.

The original authors chose a simple moving average for M_i as a practical way to account for slow-moving drift. While effective in their tests, such preprocessing may be overly

simplistic for complex real-world data. Many time series exhibit seasonality, abrupt level shifts, or non-linear trends that a plain moving average might not capture, leaving residuals X'_i still non-stationary. In this thesis, we build on the DSPOT idea but propose alternative preprocessing strategies to handle a broader class of non-stationary behaviours. The goal is to preserve the conceptual elegance of POT-based anomaly detection, ensuring that its stationarity assumptions hold more reliably in dynamic environments.

3. Methodology

3.1. Datasets

We evaluate POT-based anomaly detection methods on three real-world time series datasets sourced from the Numenta Anomaly Benchmark (NAB) repository (Lavin and Ahmad, 2015). The NAB framework is a widely used benchmark for streaming anomaly detection, providing publicly available time series with ground truth anomaly labels across a variety of domains. These datasets vary both in the nature of their non-stationarity and in the kinds of anomalies they contain. Together, they provide a strong basis for evaluating the performance of the DSPOT, and our proposed extensions. A detailed description of each dataset is provided below:

1. **CPU Utilisation (AWS EC2 Server):** This dataset records the CPU usage, of an Amazon EC2 instance at 5-minute intervals. The series displays a stable baseline with small stochastic fluctuations and periodic sharp spikes, typically lasting only one or two samples. Such spikes are attributable to short-lived workload bursts (e.g. sudden traffic surges or computational tasks). The overall distribution is piecewise stationary: it remains stable for extended periods but undergoes an abrupt downward level shift mid-series, after which the baseline settles at a lower mean. No strong daily cycle or long-term trend is visible in either segment.
2. **Twitter Volume (AMZN Hashtag):** This dataset comprises minute-by-minute counts of tweets containing the “AMZN” hashtag. The time series shows a clear 24-hour periodicity, with tweet activity increasing during business hours and dropping overnight. Superimposed on this daily cycle are occasional large spikes, often more than an order of magnitude above the baseline, corresponding to external events (e.g. news about Amazon or major announcements). The amplitude of the daily

peaks and the background variance both fluctuate substantially over time, creating a strongly non-stationary process. Importantly, anomalies here are not simply the global maxima but also include events that are extreme relative to the local daily baseline. This dataset promises to be a demanding case for anomaly detection, as naive static thresholds either over-trigger on regular daytime peaks or fail to flag subtle but meaningful surges occurring in quieter periods.

3. **Traffic Occupancy (Road Sensor 6005):** This dataset consists of 5-minute readings of freeway lane occupancy (percentage of time a loop detector registers a vehicle). A weekly cycle is evident in the alternation between these patterns. Anomalies in this dataset manifest as unusually high congestion peaks that rise well above the typical rush-hour maxima observed in the daily pattern. The dataset also contains missing data, including a substantial contiguous gap in the historical record. For our analysis, we use a cleaned, contiguous segment. The strong periodic structure induces cyclic drift in both the mean and variance, making the raw series far from stationary. This dataset is therefore an ideal stress test for our seasonality removal and normalisation procedures.

In two of the three cases (for Occupancy and Tweets), we have multiple segments of data from different time periods (with similar characteristics); we use those to repeat our experiments and average the results, to ensure that our findings are not specific to one particular timeframe.

3.2. Evaluation and Validation Strategy

To assess the performance of the POT-based anomaly detection methods, we adopt a comprehensive event-based evaluation using precision, recall, F1-score, and mean detection delay. Because anomalies in time series are often contiguous periods of abnormal behaviour (rather than single isolated points), an event-based metric is more meaningful than pointwise error rates. We define the evaluation metrics as follows:

- **True Positive (TP):** An actual anomaly event (as labelled in the dataset) that was successfully detected by the algorithm. We count an event as detected if at least one alarm was raised during the event’s duration. Multiple alerts within the same event still count as one TP in event-based evaluation.

- **False Positive (FP):** An alarm raised by the algorithm that does not correspond to any true anomaly event. These are “false alarms”: the detector signalled an anomaly at a time that was actually normal. In event terms, any alarm that falls outside the boundaries of a true event is a false positive. If the detector raises multiple alarms in a completely normal period, those are considered multiple FPs.
- **False Negative (FN):** An actual anomaly event that the detector failed to catch. This means throughout the duration of that event, the algorithm never raised an alarm.

Using the counts of these outcomes, we compute the standard metrics:

- **Precision (P):** The fraction of detected events that were true anomalies: $P = \frac{TP}{TP+FP}$, answering “when the system signals an anomaly, how often is it correct?”.
- **Recall (R):** The fraction of true anomaly events that were detected: $R = \frac{TP}{TP+FN}$. This measures the detector’s sensitivity, answering “how many of the actual anomalies did it manage to detect?”.
- **F1-score:** The harmonic mean of precision and recall: $F1 = 2 \cdot \frac{P \cdot R}{P + R}$. The F1-score provides a single balanced measure that accounts for both false alarms and missed detections.
- **Mean Detection Delay:** For each true anomaly event that is detected (TP), we calculate the time difference between the start of the event (the first timestamp of the anomaly as per ground truth) and the detector’s first alarm within that event. This is the detection delay for that event. We then average this delay across all TP events to get the mean detection delay. This metric captures the responsiveness of the detector.

Averaged Results Finally, to ensure that our results are reliable and not due to randomness, we apply our evaluation on multiple slices of data where possible (as mentioned in Section 4.1). For two of the datasets, we run the same detection experiments on those and compute the metrics, then average the performance across the runs. This helps demonstrate that a method’s performance is consistent and not over-fitted to one particular dataset instance. In the upcoming Results chapter, we will present precision, recall, F1, and delay for each method on each dataset (averaging multiple runs where

applicable), and use these to compare the effectiveness of the baseline approach versus our enhanced techniques that we discussed in detail in the sections below.

3.3. Alternative Preprocessing Pipeline, 1.5-2 pages

3.4. Declustering (not implemented yet, but I am writing the code now), 1.5-2 pages

3.5. EQD-enhanced Threshold Selection, 1.5-2 pages

Paragraphs. In the rare event that you need a more deeply nested structure within a subsubsection, the `\paragraph{}` command can be useful.

4. Results

4.1. Preprocessing effects

Here, I will describe the results for MA vs. my preprocessing pipeline; I will also include figures that will clearly show if there is an improvement in the iid assumption of the resulting residuals.

4.2. Declustering

No results yet; currently writing the code for this one.

4.3. EQD Threshold Selection

Here, I will compare whether this leads to better performance results and discuss why/why not.

4.4. Standard Detector Comparison

Wanted to conduct a comparative analysis; not sure if I have time/space for this one. If not done, it will be moved to the Future Work suggestions.

Question 1. Would you agree with this structure or suggest including the results parts after I describe them in the Methodology Chapter?

5. Discussion

Tips for the Discussion section.

- Begin your discussion with a summary and the main findings of your research in 2-5 sentences.
- Describe the implications of your main findings in the context of existing work concisely and precisely using scientific language.
- Describe the limitations in your statistical methods and your main findings. Be honest about the limitations in your approach, and substantiate what could have been done differently as needed. Explain if your main findings are robust or sensitive to these limitations.
- Avoid Subsections and Subsubsections in the Discussion.
- In the last paragraph, conclude your report with a pitch using plain language that summarises the key implications of your research in the context of previous work. Write this last paragraph for the Imperial Press Office or journalists as audience.
- Aim for approximately 1-3 pages, similar in style to a general science or statistics research paper.

6. Endmatter

Tips for the Endmatter section.

- Include a data availability statement. Examiners must be able to reproduce your analyses on all the data investigated as would be the case during peer review of a research paper.
- Include a reproducibility statement, including the URL under which your code can be found. Examiners must be able to see your code, and they must be able to run your code.

- Include a statement on how large language models or grammar tools were used in the writing of your research report, as a whole. For example, “For this report, GPT-4o was used to copy-edit introduction and conclusion text via ChatGPT graphical user interface. Grammarly was used to improve the quality of written English throughout the report. Where the output of specific queries is included in the document this is clearly quoted and cited as personal communications.”

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A. Supplementary Materials

Tips for the Supplement. Please note that the Supplementary Material is not normally read by your examiners. Include material that you might like to include for your own reference, for reference of others beyond your examiners, or as backup to your thesis in case of further discussion.

Here, we use the Supplement to provide further tips on writing your thesis.

B. Figures

Tips for the Figures. It is better to create figures in a vector-based format, such as PDF. For plots with very many elements these vector-based graphics can become large and slow to load. In these cases a high-resolution bitmap graphic, such as a png, is preferred.

Remember to make fonts in figures large enough to be read once the plots are embedded in the report (compare the two plots within Figure 1). You may need to make multiple versions of the same figure if you plan to use it in your report, poster and presentation.

All figures must be referenced in the main text of the report, in the order that they are referenced.

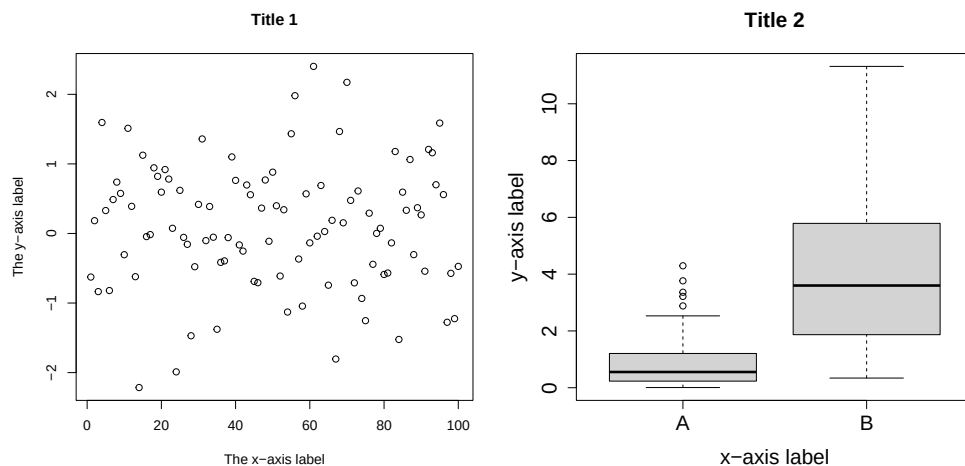


Figure 1.: An example scatterplot (left) and box plot (right).

C. Tables

Tips for the Tables. Tables can be a useful way of displaying summary statistics. An example of a simple table is presented as Table 1. All tables must be referenced in the main text of the report, in the order that they are referenced.

z	$P(Z < z)$
1.281	0.900
1.645	0.950
1.960	0.975
2.326	0.990
2.576	0.995

Table 1.: Partial table showing values of z for $P(Z < z)$, where Z has a standard normal distribution.

D. Referencing sources, sections and items

D.1. Referencing external sources

You should give attribution to the academic books and articles that you draw from and build upon in your research report. You must give full attribution if and how you used generational AI during your research, referencing its use as outlined in our Guidance to using generational AI on the MSc in Statistics.

You should use inline or parenthetical references to your sources. Inline references are used when a source is used as a noun within your sentence. For example, ? describes the bootstrap procedure. Parenthetical references are used to support a claim where the source is not explicitly mentioned within the sentence. For example, the bootstrap procedure stemmed from an earlier approach called the jackknife, which withholds a single data entry at each iteration (?).

Note that to make the references appear, you will need to compile the bibtex, otherwise you may just see question marks where the references should be. The small selection of example bibtex references used in this template are provided in the file `refs.bib`. These should be replaced with your own bibtex entries.

You can also cite works that cannot be recovered directly by the reader. This is cited as a *personal communication*, which historically would have represented an in-person

conversation or letter exchange. This now extends to many other forms of irrecoverable exchange including: email or instant messaging, interviews, live speeches, meetings and phone calls or notes made during a lecture that were not formally distributed.

You should use a personal communication citation only when a recoverable source is not available and when the communication or work contains original content that is not readily available in any other source. For example, the proprietary models currently used rely strongly on the assumption of independent observations, despite it leading to overly confident parameter estimates (?).

To reference the output of a large language model using BibTeX, you can follow a format similar to citing software or personal communication. Here's a recommended format based on academic conventions as of 2025:

```
@misc{chatgpt2025a,  
  author      = {{OpenAI}},  
  title       = {{ChatGPT response to a user query: how should I  
                reference this query using BibTeX?}},  
  year        = {2025},  
  howpublished = {\url{https://chat.openai.com}},  
  note        = {Accessed: 2025-06-24}  
}
```

This can then be referenced within the main text ([OpenAI, 2025](#)).

D.2. Referencing elements of your report

You might also want to cross reference results or claims from another part of your report such as a section, result or equation. This is done by assigning a `\label{}` to the item you wish to cross-reference and then using that label within `\ref{}`. For example, Theorem [E.2](#) is proved in Section [E](#). When referencing displayed mathematics, use `\eqref{}`. As an example, see Equation [\(6\)](#).

D.2.1. Labelling tables and figures

It is important that the label command `\label{LABELNAME}` comes **after** the caption command. See Table 1 and Figure 1 as examples of this.

D.3. Quoting sources

If you wish to quote a source, be sure to use quotation marks and cite the reference. The `\usequote` command is useful here:

“It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light, it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way - in short, the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only.” (?)

E. Definitions, theorems and examples

The following environments are supported: Definition, Theorem, Proof, Proposition, Lemma, Remark, Example.

Definition E.1. The **variance** of a random variable X is defined as

$$\text{Var}(X) = \text{E}[(X - \text{E}[X])^2]. \quad (5)$$

Theorem E.2. Given a random variable X , over all values $a \in \mathbb{R}$,

$$\min_{a \in \mathbb{R}} \text{E}[(X - a)^2] = \text{E}[(X - \text{E}[X])^2]. \quad (6)$$

Proof. Starting with the left-hand side of Equation (6),

$$\begin{aligned}
 \mathbb{E}[(X - a)^2] &= \mathbb{E}[(X - \mathbb{E}[X] + \mathbb{E}[X] - a)^2] \\
 &= \mathbb{E}[(X - \mathbb{E}[X])^2] + 2\mathbb{E}[(X - \mathbb{E}[X])(\mathbb{E}[X] - a)] + \mathbb{E}[(\mathbb{E}[X] - a)^2] \\
 &= \mathbb{E}[(X - \mathbb{E}[X])^2] + (\mathbb{E}[X] - a)^2 \\
 &\geq \mathbb{E}[(X - \mathbb{E}[X])^2],
 \end{aligned}$$

since $\mathbb{E}[X]$ is a real number and $(\mathbb{E}[X] - a)^2 \geq 0$, and the third line follows from linearity of expectation:

$$\mathbb{E}[(X - \mathbb{E}[X])(\mathbb{E}[X] - a)] = (\mathbb{E}[X] - a) \mathbb{E}[(X - \mathbb{E}[X])] = (\mathbb{E}[X] - a) (\mathbb{E}[X] - \mathbb{E}[X]) = 0,$$

since $\mathbb{E}[\mathbb{E}[X]] = \mathbb{E}[X]$, which proves the result. $\square$

Remark E.3. This theorem shows that the minimum of the quantity $\mathbb{E}[(X - a)^2]$ is equal to $\text{Var}(X)$. In some sense, this makes the variance a natural measure of dispersion if we are taking the metric to be the squared deviation of X .

Lemma E.4 (Stein’s Lemma). Let $X \sim N(\mu, \sigma^2)$, and let g be a differentiable function satisfying $\mathbb{E}[|g'(X)|] < \infty$. Then

$$\mathbb{E}[g(X)(X - \mu)] = \sigma^2 \mathbb{E}[g'(X)].$$

Proposition E.5 (Popoviciu’s inequality). Suppose that the random variable X is known to only take values in the bounded range $[a, b]$. Then

$$\text{Var}[X] \leq \frac{(b - a)^2}{4}.$$

Example E.6. Suppose $X \sim \text{Bern}(p)$, for some $p \in [0, 1]$. Then, since $X \in \{0, 1\}$, X is bounded between 0 and 1 and so $\text{Var}[X] \leq \frac{1}{4}$.

Note that Equations (5) and (6) are numbered because they are referred to elsewhere in the main text. All other equations are unnumbered.

Each of these environments may be labelled and referenced as with sections. For example, we might discuss Definition E.1 or Theorem E.2 in this paragraph.